Predictions

- Problem setup
- Linear regression

Prediction problems

We have input/output data

$$\mathcal{D} = \left\{(x_n, y_n)\right\}_{n=1}^{N} \subset \mathbb{R}^K \times \mathbb{R}$$

and seek a regression function $f : \mathbb{R}^K \mapsto \mathbb{R}$

In linear regression we apply a feature expansion

$$\phi : \mathbb{R}^K \mapsto \mathbb{R}^{1\times M}$$

and seek only solutions of the form

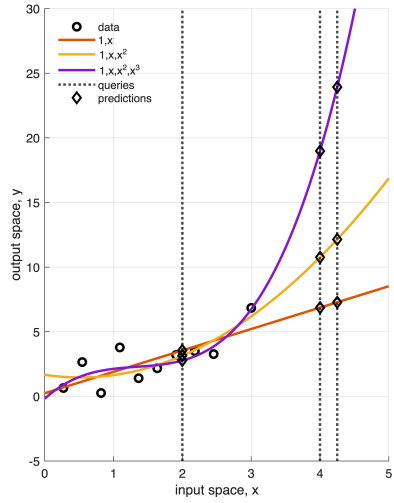
$$f_c(x) = \phi(x)c$$

for appropriate coefficients $c \in \mathbb{R}^M$

- Our goal is to obtain predictions of the targets

$$y^* \in \mathbb{R}$$

at given queries $x^* \in \mathbb{R}^K$



Bayesian prediction problems

We consider data $\left\{(x_n, y_n)\right\}_{n=1}^{N} \subset \mathbb{R}^K \times \mathbb{R}$, a function $\phi : \mathbb{R}^K \mapsto \mathbb{R}^{1\times M}$, and seek solutions

$$f_c(x) = \phi(x)c$$

- Our basic Bayesian model consists of

$$\begin{aligned} c &\sim \text{Normal}_M(\mu, C) \\ y_n|c &\sim \text{Normal}(f_c(x_n), \eta), \qquad n = 1, \ldots, N \end{aligned}$$

- To allow for predictions y^* at a query x^* we consider an extended model

$$\begin{aligned} c &\sim \text{Normal}_M(\mu, C) \\ y_n|c &\sim \text{Normal}(f_c(x_n), \eta), \qquad n = 1, \ldots, N \\ y^*|c &\sim \text{Normal}(f_c(x^*), \eta) \end{aligned}$$

Our predictive distribution is

$$p(y^*|y_{1:N}) = \int dc\, p(y^*, c|y_{1:N}) = \int dc\, p(y^*|c, y_{1:N})\, p(c|y_{1:N}) = \int dc\, p(y^*|c)\, p(c|y_{1:N})$$

Bayesian prediction problems

We consider data $\left\{(x_n, y_n)\right\}_{n=1}^{N} \subset \mathbb{R}^K \times \mathbb{R}$, a function $\phi : \mathbb{R}^K \mapsto \mathbb{R}^{1\times M}$, and seek solutions

$$f_c(x) = \phi(x)c$$

- To allow for predictions y_ℓ^* at multiple queries x_ℓ^* we consider an extended model

$$\begin{aligned} c &\sim \text{Normal}_M(\mu, C) \\ y_n|c &\sim \text{Normal}\left(f_c(x_n), \eta\right), && n = 1, \ldots, N \\ y_\ell^*|c &\sim \text{Normal}\left(f_c(x_\ell^*), \eta\right), && \ell = 1, \ldots, L \end{aligned}$$

Our predictive distribution is

$$\begin{aligned} p\left(y_{1:L}^*|y_{1:N}\right) = \int dc\, p\left(y_{1:L}^*, c|y_{1:N}\right) &= \int dc\, p\left(y_{1:L}^*|c, y_{1:N}\right) p\left(c|y_{1:N}\right) \\ &= \int dc\, p\left(y_{1:L}^*|c\right) p\left(c|y_{1:N}\right) \end{aligned}$$

The predictive distribution

Our extended model consists of

$$
\begin{aligned}
c &\sim \text{Normal}_M(\mu, C) \\
y_n | c &\sim \text{Normal}(f_c(x_n), \eta), && n = 1, \ldots, N \\
y^*_\ell | c &\sim \text{Normal}(f_c(x^*_\ell), \eta), && \ell = 1, \ldots, L
\end{aligned}
$$

This is equivalent to the model

$$
\begin{aligned}
c &\sim \text{Normal}_M(\mu, C) \\
\begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix} \Bigg| c &\sim \text{Normal}_N\left(\begin{bmatrix} \phi(x_1) \\ \vdots \\ \phi(x_N) \end{bmatrix} c, \eta \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix} \right) \\
\begin{pmatrix} y^*_1 \\ \vdots \\ y^*_L \end{pmatrix} \Bigg| c &\sim \text{Normal}_L\left(\begin{bmatrix} \phi(x^*_1) \\ \vdots \\ \phi(x^*_L) \end{bmatrix} c, \eta \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix} \right)
\end{aligned}
$$

$$
\begin{aligned}
c &\sim \text{Normal}_M(\mu, C) \\
Y | c &\sim \text{Normal}_N(\Phi c, \eta I_N) \\
Y^* | c &\sim \text{Normal}_L(\Phi^* c, \eta I_L)
\end{aligned}
$$

The predictive distribution

Our extended model consists of

$$
\begin{aligned}
c &\sim \text{Normal}_M\left(\mu, C\right) \\
Y|c &\sim \text{Normal}_N\left(\Phi c, \eta I_N\right) \\
Y^*|c &\sim \text{Normal}_L\left(\Phi^* c, \eta I_L\right)
\end{aligned}
$$

This model leads to

$$
c|Y \sim \text{Normal}_M\left(\mu', C'\right) \qquad \mu' = C'\left(C^{-1}\mu + \frac{1}{\eta}\Phi^t Y\right) \qquad C' = \left(C^{-1} + \frac{1}{\eta}\Phi^t\Phi\right)^{-1}
$$

$$
Y^*|c \sim \text{Normal}_L\left(\Phi^* c, \eta I_L\right)
$$

Our predictive distribution is

$$
\begin{aligned}
p\left(Y^*|Y\right) = \int dc\, p\left(Y^*, c|Y\right) &= \int dc\, p\left(Y^*|c, Y\right) p\left(c|Y\right) \\
&= \int dc\, p\left(Y^*|c\right) p\left(c|Y\right) \\
&= \int dc\, \text{Normal}_L\left(Y^*; \Phi^* c, \eta I_L\right) \text{Normal}_M\left(c; \mu', C'\right) = \text{Normal}_L\left(Y^*; \mu'', C''\right)
\end{aligned}
$$

The predictive distribution

Our extended model consists of

$$
\begin{aligned}
c &\sim \text{Normal}_M\left(\mu, C\right) \\
Y|c &\sim \text{Normal}_N\left(\Phi c, \eta I_N\right) \\
Y^*|c &\sim \text{Normal}_L\left(\Phi^* c, \eta I_L\right)
\end{aligned}
$$

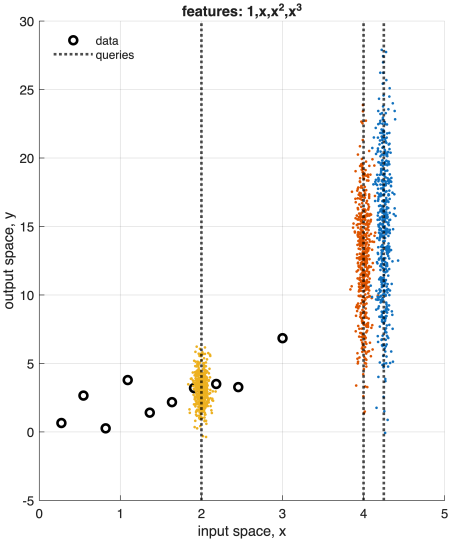
This model leads to

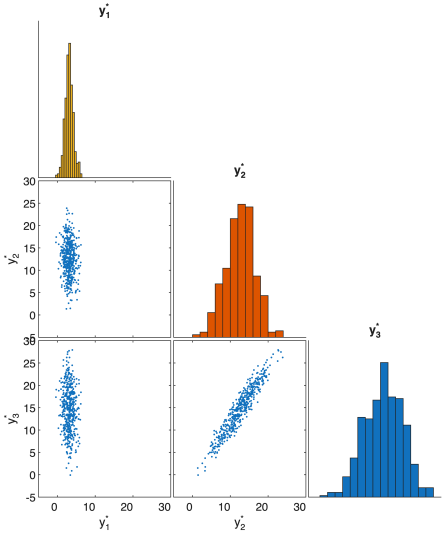
$$
\begin{aligned}
c|Y &\sim \text{Normal}_M\left(\mu', C'\right) \qquad \mu' = C'\left(C^{-1}\mu + \frac{1}{\eta}\Phi^t Y\right) \qquad C' = \left(C^{-1} + \frac{1}{\eta}\Phi^t\Phi\right)^{-1} \\
Y^*|c &\sim \text{Normal}_L\left(\Phi^* c, \eta I_L\right)
\end{aligned}
$$

which, in turn, leads to

$$
Y^*|Y \sim \text{Normal}_L\left(\mu'', C''\right) \qquad \mu'' = \Phi^*\mu' \qquad C'' = \Phi^* C'(\Phi^*)^t + \eta I_L
$$

Example


features: 1,x,x²,x³
data
queries
output space, y
input space, x
30
25
20
15
10
5
0
-5
0
1
2
3
4
5


y*1
y*2
y*3
30
25
20
15
10
5
0
-5
0
10
20
30

Bayesian prediction problems under model uncertainty

We have a pool with K models, each characterized by potentially different sizes $M_{1:K}$, feature expansions $\phi_{1:K}(x)$, and coefficients $c_{1:K}$

Combining all models in the pool, we obtain a master model

$$\begin{aligned} \kappa &\sim \text{Categorical}_{1:K}\left(\pi_1, \ldots, \pi_K\right) \\ c_k &\sim \text{Normal}_{M_k}\left(\mu_k, C_k\right), && k = 1, \ldots, K \\ y_n | \kappa, c_{1:K} &\sim \text{Normal}\left(\phi_\kappa(x_n) c_\kappa, \zeta\right), && n = 1, \ldots, N \end{aligned}$$

- To allow for predictions y_ℓ^* at queries x_ℓ^* we consider an extended master model

$$\begin{aligned} \kappa &\sim \text{Categorical}_{1:K}\left(\pi_1, \ldots, \pi_K\right) \\ c_k &\sim \text{Normal}_{M_k}\left(\mu_k, C_k\right), && k = 1, \ldots, K \\ y_n | \kappa, c_{1:K} &\sim \text{Normal}\left(\phi_\kappa(x_n) c_\kappa, \zeta\right), && n = 1, \ldots, N \\ y_\ell^* | \kappa, c_{1:K} &\sim \text{Normal}\left(\phi_\kappa(x_\ell^*) c_\kappa, \zeta\right), && \ell = 1, \ldots, L \end{aligned}$$

Bayesian prediction problems under model uncertainty

Our extended master model consists of

$$
\begin{aligned}
\kappa &\sim \text{Categorical}_{1:K}\left(\pi_1, \ldots, \pi_K\right) \\
c_k &\sim \text{Normal}_{M_k}\left(\mu_k, C_k\right), && k = 1, \ldots, K \\
y_n|\kappa, c_{1:K} &\sim \text{Normal}\left(\phi_\kappa(x_n)c_\kappa, \zeta\right), && n = 1, \ldots, N \\
y^*_\ell|\kappa, c_{1:K} &\sim \text{Normal}\left(\phi_\kappa(x^*_\ell)c_\kappa, \zeta\right), && \ell = 1, \ldots, L
\end{aligned}
$$

Our predictive distribution is

$$
\begin{aligned}
p\left(y^*_{1:L}|y_{1:N}\right) &= \sum_{\kappa=1}^{K} p\left(y^*_{1:L}, \kappa|y_{1:N}\right) \\
&= \underbrace{\sum_{\kappa=1}^{K} \underbrace{p\left(y^*_{1:L}|y_{1:N}, \kappa\right)}_{\substack{\kappa^{\text{th}} \text{ model} \\ \text{predictions}}} \underbrace{p\left(\kappa|y_{1:N}\right)}_{\substack{\kappa^{\text{th}} \text{ model} \\ \text{selection}}}}_{\text{model averaging}}
\end{aligned}
$$

This way, our predictions, are averaged over all models in the pool ...posterior mixture predictions